



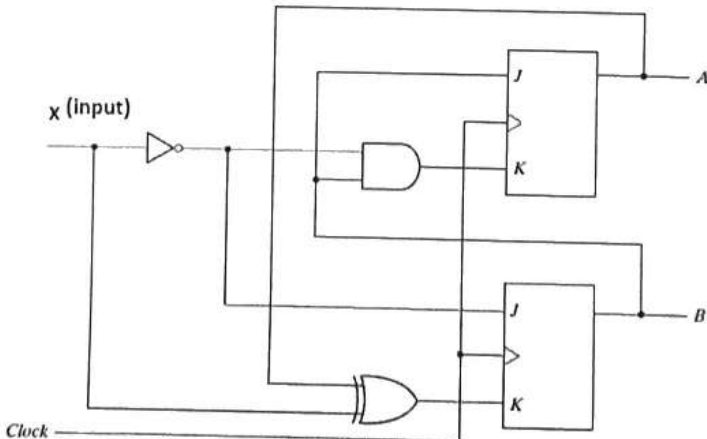
SOMAIYA
VIDYAVIHAR UNIVERSITY

Semester: January 2024 –April 2024		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 03	Class: SY	Semester: _III (SVU 2020)
Programme: BTech EXTC	BTech EXTC	
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: EXTC
Course Code: 116U03C303	Name of the Course: Digital Logic Design	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks	CO	BT
Q1	Solve any Four	20		
i)	Perform the following operations. Show your calculations 1. Convert $(A20)_{16}$ into decimal. 2. $(543)_{10}$ into BCD	5	CO1	AP
ii)	Evaluate $(78)_{10} - (53)_{10}$ using two's complement method	5	CO1	AP
iii)	Prove the expression $(AB + C + D)(\bar{C} + D)(\bar{C} + D + E) = AB\bar{C} + D$	5	CO2	AP
iv)	Design Half adder using basic gates.	5	CO3	AP
v)	Draw JK flip flop using gates. Write truth table for the same.	5	CO4	RE
vi)	Differentiate between Moore and Melay sequential machines.	5	CO4	UN

Que No.	Question	Max. Marks	CO	BT
Q2 A	Solve the following	10		
i)	Design a full adder using 3 to 8 line decoder.	5	CO3	AP
ii)	Write truth table for a 2 bit comparator.	5	CO3	AP
	OR			
Q2 A	If $Y(A,B,C,D) = AB\bar{C} + D$ 1. Draw the corresponding K map represented by above expression 2. Write canonical SOP form of expression 3. Realise using NAND gates only	10	CO2	AP
Q2 B	Solve any One	10		
i)	Use Quine – Mc Cluskey to minimize the following expression $F(ABCD) = \sum m(0, 2, 4, 6, 7, 8, 9, 10, 13)$	10	CO2	AP
ii)	If $Y(A, B, C, D) = \sum m(0, 1, 4, 5, 8, 9, 10, 12, 13, 14)$ I. Express the above expression in minimized the SOP form K – Maps II. Draw the logic diagram for the above expression using only NAND gates	10	CO2	AP

Que. No.	Question	Max. Marks	CO	BT
Q3	Solve any Two	20		
i)	Design a mod 6 synchronous counter using JK flip flops	10	CO4	AP
ii)	Design following function using suitable size PAL(Programmable Array Logic) $F1 = \sum m(0, 2, 4, 5, 9, 11)$ $F2 = \sum m(4, 5, 13, 14, 15)$	10	CO4	AP
iii)	Write truth tables and excitation tables for SR, JK and D flip flops	10	CO4	UN

Que. No.	Question	Max. Marks	CO	BT
Q4	Solve any Two	20		
i)	Draw and explain 3 bit left shift register using D flip flops. Write the truth table assuming input sequence of 1, 1, 0, 1, 0, 1, 0	10	CO4	AP
ii)	Referring to the diagram of synchronous sequential circuit below analyse the circuit. Write PS-NS table and draw state diagram.	10	CO4	AP
				
iii)	Design mod 5 Asynchronous up counter using T flip flops	10	CO4	AP

Que. No.	Question	Max. Marks	CO	BT
Q5	(Write notes / Short question type) on any four	20		
i)	Write short note on weighted and non-weighted codes	5	CO1	UN
ii)	In radix (base) 5 number system a number is expressed as (413). What is its decimal and binary equivalent?	5	CO1	AP
iii)	Write short note on Dual slope ADC	5	CO5	AN
iv)	Compare TTL and CMOS logic families (4-5 points)	5	CO1	UN
v)	$F = \sum m(0, 2, 4, 5)$. Realise using 8:1 mux	5	CO3	AP
vi)	Compare synchronous and asynchronous counter.	5	CO4	AP

Q1 (i)

$$(A20)_{16} = (2592)_{10}$$

$$(543)_{10} = \underline{0101} \underline{0100} (0011)_{BCD}$$

(3)

(2)

$$(ii) (78-53)_{10} = (25)_{10}$$

$$10001110 - 0110101$$

↓ 2's compl.

$$(1001011)$$

$$+ 1001110$$

$$\boxed{1}0011001$$

discarded

$$= (25)_{10} \rightarrow (2)$$

(2)

(1)

$$(iii) (AB+C+D)(\bar{C}+D) + (\bar{C}+D+E)$$

$$= (AB+C+D)(\underbrace{\bar{C} + \bar{C}D + \bar{C}E + \bar{C}D}_{\bar{C}} + \underbrace{D + DE}_{D})$$

$$= (AB+C+D)(\bar{C}+D)$$

$$= AB\bar{C} + ABD + 0 + CD + \bar{C}D + D$$

$$= AB\bar{C} + D(AB+C+\bar{C}+1) = \cancel{AB} + AB\bar{C} + D$$

(iv) Half adder

$$\text{Sum} = A\bar{B} + A\bar{B}$$

$$\text{Carry} = AB$$

Truth Table →

(1)

equations →

(2)

Diagram →

(2)

(v)

Diagram →

(2)

Truth Table →

(3)

Q2A) i) full adder Truth Table $\rightarrow$ (2).

$$\text{sum} = \sum m(1, 2, 4, 7) \rightarrow (1)$$

$$\text{carry out} = \sum m(3, 5, 6, 7)$$

Diagram using ~~NAND~~ decoder $\rightarrow$ (2).

ii) A B A > B A < B A = B $\rightarrow$ (5).

Q2A] $Y = AB\bar{C} + D.$

AB \ CD	00	01	11	10
00			1	
01	1	1	1	1
11	1	1	1	1
10				

$$Y = \sum m(1, 3, 5, 7, 9, 11, 12, 13, 15)$$

K map $\rightarrow$

3 marks.

SOP $\rightarrow$

3 marks

Realization $\rightarrow$

4 marks.

Q2B]. $\sum m(0, 2, 4, 6, 7, 8, 9, 10, 13).$

$$Y = A\bar{D} + \bar{B}\bar{D} + \bar{A}BC + A\bar{C}D$$

OR.

$Y = \bar{C} + A\bar{D}.$

AB \ CD	00	01	11	10
00	1	1	0	0
01	1	1	0	0
11	1	1	0	1
10	1	1	0	1

K map drawing $\rightarrow$ 3
 solving $\rightarrow$ 4
 Realization $\rightarrow$ 4

Q3. # =
i) mod 6 synchronous counter

Truth Table $\rightarrow$ 2

K-maps $\rightarrow$ 4

equations $\rightarrow$ 2

diagram $\rightarrow$ 2.

$$J_2 = Q_1 Q_0$$

$$K_2 = Q_0$$

$$J_1 = \bar{Q}_2 Q_0$$

$$K_1 = Q_0$$

$$J_0 = 1$$

$$K_0 = 1$$

ii)

$$F_1 = \bar{A}\bar{B}\bar{D} + \bar{A}B\bar{C} + A\bar{B}D \quad \left. \vphantom{\begin{matrix} F_1 \\ F_2 \end{matrix}} \right\} \rightarrow 6 \text{ marks.}$$

$$F_2 = \bar{A}\bar{B}\bar{C} + B\bar{C}D + ABC$$

PAL diagram - 4 marks

iii)

Truth Table $\rightarrow$ 6 marks

Excitation Table $\rightarrow$ 4 marks.

Q4. Diagram $\rightarrow$ 6 marks

